Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/12/2005
PI Name:	Beachy, Roger N. Ph.D.		
Project Title:	Development and Plant Gene Switch System for Expression of Multiple Genes		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	63132	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	6	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Original end date was 11/30/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC29G (jointly funded by JSC and ARC)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	According to the Report of the Workshop on Biology-based Technology to Enhance Human Well-being and Function in Extended Space Exploration, (1997), to enable long-term missions without resupply support, ALS systems must be developed to fully recycle air and water, recover resources from solid waste, grow plants for food, and process raw plant products into nutritious and palatable foods . Thus, plants will be essential for human well-being in extended space exploration1. Genetic engineering rather than conventional plant breeding methods will be necessary to develop plants used in extended space missions1. Target plants space missions include wheat, rice, white potatoes, sweet potatoes, soybeans, lettuce, peanuts and sugar beets1. These crops provide important nutrients and can be genetically engineered1. Desirable plant traits for space missions include altered plant size/form to conform to space limitations, miniature roots for hydroponic growth and reduced waste, enhanced productivity under low light and high CO2, resistance to disease, capability for early flowering, reduced lignin content, production of biomaterials such as altered oils and starches, modified metabolic pathways, and increased yield and productivity. It may also be possible to engineer multipurpose plants that could provide complete dietary requirements for humans1. To fully exploit the potential of genetic engineering, technologies to control plant gene expression under conditions of space exploration and to create multipurpose plants, e.g. gene switches, need to be developed1. To address the short-term and long-term needs of the NASA?s ALS program on plant modification outlined above, six target-focused projects are outlined below which will provide an understanding of some of the basic components of plant bioengineering listed herein and will create the foundation for modification of plants for dual use as part of ALS systems and for Earth-based applications in human and plant health and nutrition. Note: Figures are presented in Appendix A for reference purposes only.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plant modification will provide an understanding of some of the basic components of plant bioengineering and will create the foundation for modification of plants for Earth-based applications in human and plant health and nutrition.
Task Progress:	The application described four (4) major goals for the project: here we briefly describe the progress on each goal, as relevant. Thus far, we have used the gene switch system only in Arabidopsis thaliana and studies are relevant only on this species. Goal 1. Determine the length of time between addition of ligand and induction of gene expression in plants grown in soil. More than 170 different transgenic lines were developed to carry the VGE receptor and luciferase reporter gene to which the receptor binds. Of these, more than 50% showed induction after soil drenching with the ligand. (The ligand for activating gene expression in this system is methoxyfenozide.) Gene induction was observed in leaves within 30 min of addition of the soil drench, and continued to increase for more than 100 hrs, after which the level of luciferase declined. We attribute the decline in luciferase to result from growth of the plant and dilution of the ligand in the plant. The results of these studies and others associated with this goal (e.g., Goal 2) are currently being compiled for publication (Fig. 1; Z. Zhang, S. Dai, and R.N. Beachy, in preparation). Goal 2. Determine the length of time that the ligand retains activity. For this study we developed a hydroponics system for Arabidopsis and the ligand was added to the medium (plant line was like that described above, Goal 1) for approximately 24 hrs. The medium was then changed and gene expression was followed for the next 96 hrs. Induction of gene expression in hydroponic conditions was recorded within 30 min of adding the ligand, and continued for approximately 72 hrs, after which the level of luciferase declined. These studies demonstrate that it is not possible to quench the effect of the gene switch by changing the medium in which plants are grown; reducing expression of the transgene appears to rely on dilution of the ligand that will occur as a consequence of plant growth. (Fig. 2; Z. Zhang, S. Dai, and R.N. Beachy, in preparation). Goal 3. Test GS technologies for tissue specific expression of genes. Control of tissue specific gene expression has not been adequately addressed in plants; such studies are necessary in order to understand the nature of ligand penetration or transport. We initiated studies to address these and related questions in Arabidopsis. In this and related studies, the VGE chimeric receptor was under the regulation of different tissue specific promoters and the reporter gene (GUS or GFP) was under the control of a chimeric promoter with the DNA binding site of VGE. Our initial study was conducted with the PAL promoter of bean phenylalanine ammonia-lyase gene that confers specific gene expression in vascular tissues of transgenic monocot and dicot plants and the E promoter of rice tungro bacilliform virus that confers phloem specific gene expression in the plant vascular tissues. Our data demonstrate that the tissue specifically expressed VGE under the regulation of either PAL or E promoter can activate the GUS reporter gene expression in those specific tissue types of the transgenic plants only when the ligand was applied. We developed additional constructs with the VGE receptor regulated by other tissue specific promoters and transgenic plants are currently being developed. The table below lists the gene constructs that were developed: for expression in green tissues, the promoter of the chlorophyll a/b binding protein was used (Mitra et al., 1989); for expression in vascular tissues, the full-length and truncated promoter of the rice tungro bacilliform virus (Bhattacharyya-Pakrasi et al., 1993) and the promoter of bean phenylalanine ammonia-lyase gene (Leyva et al., 1992) were used; for expression in cotyledons of developing seeds, the beta-conglycinin (Chen et al., 1986) and oleosin (Lee and Huang, 1994) promoters were used; and for expression in roots, a promoter of the acidic chitinase (Samac and Shah, 1991; Samac et al., 1990) was used. Transgenic plants with the GUS reporter gene regulated by these tissue specific promoters were also produced as controls. Table 1. List of control gene constructs with the GUS (uidA) reporter gene directly regulated by different tissue specific promoters: TISSUE EXPRESSION CONSTRUCT Vascular tissue (phloem) RTBV:uidA Green tissue Cab3:uidA Immature cotyledons B-conglycinin:uidA Cotyledons Oleosin:uidA Root tissues Acidic Chitinase:uidA Table 2. Constructs to induce expression of the uidA reporter gene via the VGE receptor. TISSUE EXPRESSION CONSTRUCT Vascular tissue (phloem) RTBV:VGE/5xG-35S:uidA Vascular tissue (phloem) RTBV-E:VGE/5xG-35S:uidA Vascular tissue (phloem) PAL:VGE/5xG-35S:uidA Green tissue Cab3:VGE/5xG-35S:uidA Cotyledons of seeds B-conglycinin:VGE/5xG-35S:uidA Cotyledons of seeds Oleosin:VGE/5xG-35S:uidA Root Acidic Chitinase:VGE/5xG-35S:uidA control 5xG-35S:uidA

	Tissue specific induction of gene expression in transgenic Arabidopsis plants with listed gene constructs will be tested. This work was conducted by Z. Zhang, S. Dai and I. Ordiz. Goal 4. Develop strategies for expression of multiple genes by a single gene switch ligand. The goal of this study is to regulate the expression of multiple genes in Arabidopsis by activating expression of a transcription factor that in turn regulates multiple other genes that are controlled by the transcription factor. To date, we conducted a study in which the VGE receptor was expressed under the control of CsVMV constitutive promoter, and the RF2a gene was regulated by a fusion promoter that contained the VGE DNA binding site. The GUS reporter gene was under the control of the RTBV promoter which contains the DNA binding site for RF2a. Without application of ligand, the expression of GUS gene in transgenic plants was limited to vascular tissues, consistent with the anticipated expression pattern of the RTBV promoter. However, the expression pattern of the reporter was constitutive when the ligand was applied. This result indicates that, upon ligand binding, VGE activates the expression of RF2a in all cell types. In turn, the expressed RF2a in these tissues activates the expression of GUS reporter in these cells. This work was completed by Z. Zhang and S. Dai. This goal will be substantially enhanced during years 2 and 3 by J. Yang.
Bibliography Type:	Description: (Last Updated: 02/27/2004)